PER SCHOLAS

Report

CAP 931: Sales Agent Prototype Using Multi-Agent GPT Models

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Overview

This capstone produced a Streamlit sales assistant prototype that uses three sequential GPT agent roles—Research, Account Analyst, and Brief Writer—to create source-linked sales account briefs. The prototype accepts product details, a prospective company and its URL, competitor URLs, and optional information. In the test run, it researched wooden baseball bats for Target Corporation and generated PDF and Markdown reports. It identified relevant retail listings, named relevant Target executives, and marked unsupported claims and unknown information. All 10 automated tests passed.

Technical Design and Model Choice

The Streamlit interface in `app.py` collects the product and prospective-account details. It validates required fields and public URLs, and it can extract text from optional TXT, Markdown, PDF, or DOCX product documents.

`src/sales_agent/pipeline.py` runs three sequential roles through the OpenAI Responses API. The Research Agent uses web search to gather public information and source links. The Account Analyst works from that research packet to identify evidence, inferences, and gaps. The Brief Writer creates the concise account brief. The other modules handle URL validation, document extraction, and PDF and Markdown report generation.

The runs used `gpt-6-luna`, the configured default model, for all three roles. Keeping the model consistent made the input-correction runs comparable. This project did not benchmark different models, costs, or response times, so it does not claim that this model is the best choice.

Steps

1. Opened the CAP 931 project in VS Code on macOS.
2. Set up the Python environment and configured the API key in `.env`.
3. Started the Streamlit app with `uv run streamlit run app.py`.
4. Entered the wooden baseball bat product details, Target Corporation as the prospect, and Marucci, Mizuno, and Victus as competitor references. Left the optional contact and product overview blank.
5. Ran the account research and reviewed the generated brief, including its sources, unsupported claims, and unknowns.



Sales Account Brief: Target Corporation

Research date: 2026-10-12 | Product: Base Ball Bats (Sport) | Model: gpt-6-luna

Company: https://corporate.target.com/

Account snapshot

- Account: Target Corporation; product: wooden baseball bats; contact: not provided.

- Proposed value proposition: "The best wooden bat in the market." This claim is not substantiated in the reviewed packet.

Strategy signals

- Target describes priorities including differentiated assortments, value, digital discovery and in-store experiences; its materials also identify youth sports essentials as suited to in-store shopping. (2026 strategy (https://corporate.target.com/news-features/article/2026/03/target-growth-strategy-2026?utm_source=openai), 2025 annual report (https://corporate.target.com/investors/annual/2025-annual-report?utm_source=openai))

- Inference: A wooden-bat proposal may fit Target's existing category and youth-sports assortment. The sources do not establish a specific priority for wooden bats or premium performance.

Competitor mentions

- Target's wooden-bat listings include Marucci and Mizuno; a Victus listing identifies Smash It Sports as the seller and says the item is sold and shipped by that third party. (Wood bats (https://www.target.com/c/baseball-bats-equipment-gear-sports-outdoors/wood/-/N-pk1lmZ5y6vh?utm_source=openai), Victus bats (https://www.target.com/c/baseball-bats-equipment-gear-sports-outdoors/victus/-/N-pk1lmZkqb4d?utm_source=openai))

- These are retail-listing mentions, not evidence of strategic competitor status or a corporate relationship.

Relevant leadership

- Cara Sylvester, EVP and Chief Merchandising Officer: Leads assortment curation and merchandising; the most directly relevant named executive for an assortment proposal. (Bio (https://corporate.target.com/about/leadership-team/cara-sylvester?utm_source=openai))

- Michael Fiddelke, CEO: Associated with Target's broader merchandising, guest-experience and technology agenda. (Bio (https://corporate.target.com/about/leadership-team/michael-fiddelke?utm_source=openai))

- Prat Vemana, EVP and Chief Information and Product Officer: Leads technology, data, AI and user experience; not identified as the bat-category buyer. (Bio (https://corporate.target.com/about/leadership-team/prat-vemana?utm_source=openai))

- Category buyer not found in the reviewed sources.

Suggested sales angle

- Inference: Present product differentiation, value, accurate product data, and dependable inventory and fulfillment as potential points of relevance to Target's stated priorities. These are not published vendor requirements.

- Avoid asserting "best" without substantiation: the packet does not show that Target requires or uses this claim.

Unknowns to verify

- Target's wooden-bat requirements, interest in premium performance, buying process, and criteria for substantiating a "best" claim: Not found in the reviewed sources.

- Category buyer and target contact: Not found in the reviewed sources.

Sources

Target Outlines Strategic Plan for a New Chapter of Growth in 2026 and Beyond:
https://corporate.target.com/news-features/article/2026/03/target-growth-strategy-2026?utm_source=openai

Wood : Baseball Bats : Target:
https://www.target.com/c/baseball-bats-equipment-gear-sports-outdoors/wood/-/N-pk1lmZ5y6vh?utm_source=openai

Target Plus Fuels Discovery with More Brands and Newness:
https://corporate.target.com/news-features/article/2026/07/target-plus-growth?utm_source=openai

Victus : Baseball Bats : Target:
https://www.target.com/c/baseball-bats-equipment-gear-sports-outdoors/victus/-/N-pk1lmZkqb4d?utm_source=openai

How Target Technology Is Powering Retail Growth:
https://corporate.target.com/news-features/article/2025/10/technology-powering-growth?utm_source=openai

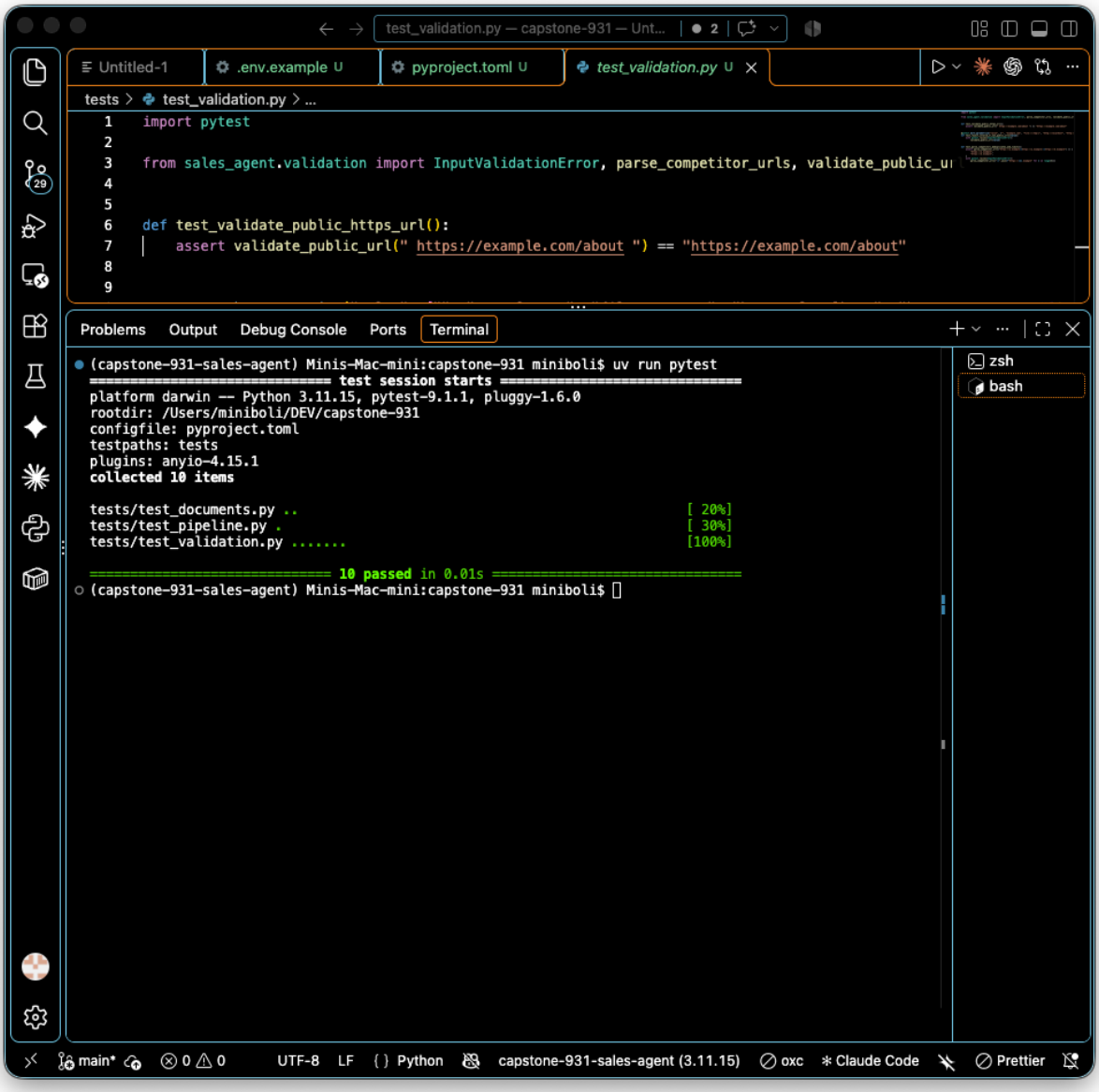
2025 Annual Report | Target Corporate: https://corporate.target.com/investors/annual/2025-annual-report?utm_source=openai

Michael Fiddelke bio | Target Corporation: https://corporate.target.com/about/leadership-team/michael-fiddelke?utm_source=openai

Cara Sylvester bio | Target Corporation: https://corporate.target.com/about/leadership-team/cara-sylvester?utm_source=openai

6. Saved the generated PDF and Markdown brief in `outputs/`.

7. Ran `uv run pytest`; all 10 tests passed.



Challenges and Solutions

- The first competitor references, Snowflake and Databricks, did not match the baseball-bat product. I replaced them with Marucci, Mizuno, and Victus and ran the research again.
- The phrase “best wooden bat” could not be verified from the sources. The updated brief labels it as an unsubstantiated seller claim instead of presenting it as fact.
- The research did not identify Target’s baseball-bat category buyer. The report records this as unknown rather than guessing.
- The first brief’s suggested sales-angle text ended abruptly. I reviewed the updated brief and used the complete version.

Production Deployment and Future Improvements

This prototype runs locally and has not been deployed publicly. A production version could run on a managed hosting service with HTTPS and user authentication. The API key should be stored in the host’s secrets manager, not in the application code. Upload limits, request limits, and cost monitoring should help protect the service. Maintenance would include updating dependencies, rerunning tests, and reviewing source quality and report accuracy.

Future improvements could check that every claim has a source, flag incomplete reports before download, and compare prompt or model choices using the same inputs. An optional alert feature could notify sales representatives when new public information matches selected keywords. These are proposed enhancements; they are not implemented in this prototype.

Conclusion

The prototype successfully generated a source-linked account brief for the wooden baseball bat example, and all 10 automated tests passed. The research found possible alignment with Target's sports assortment and identified relevant retail listings, but it did not establish that Target is prioritizing wooden bats or identify the category buyer. The product's "best wooden bat" statement remains an unverified seller claim. The brief is useful for preparing research questions, but its sources and claims should be checked before making a sales approach.

Appendix

- **Generated account brief:** PDF and Markdown files in `outputs/`.
- **Evidence screenshots:** generated brief and passing tests in `docs/evidence/`.
- **Application:** `app.py` and the files in `src/sales_agent/`.
- **Automated tests:** files in `tests/`; all 10 passed.
  - **Time log:** `docs/TIME_LOG.csv` — retrospective estimates.
  - **Experiment log:** `docs/EXPERIMENT_LOG.md` — two runs; single-pass baseline not performed.
- Keep your `.env` file and API key out of the report and submission.